

**Owner-approved ExamSIDE question-demand
slice for plane motion, now semantically:
Construction**

Split one plane-motion event into independent x/y components with one clock

You need

- Can read signed vector components in a declared x-y frame.
- Choosing axes and keeping components signed (Physics)

The key step

Perpendicular components may be solved separately because each axis equation contains only that axis's quantities, but they still describe one object at one physical time.

Solve each axis on its own, but read both coordinates at one shared instant before calling the pair a position.

1. **Choose one fixed x-y frame, one origin and one time origin for the whole plane-motion event.**

Why valid: A component value is meaningful only after its frame and sign convention are fixed.

Result: one declared frame, origin, axis orientation and $t=0$

2. Represent position, velocity and acceleration as signed x and y components in that same frame.

Why valid: A plane vector is completely represented by its perpendicular Cartesian components once the frame is declared.

Result: For $x = 3t$ and $y = 5t - t^2$ at $t = 1$ s: $(x, y) = (3, 4)$ m, $(v_x, v_y) = (3, 3)$ m/s, $(a_x, a_y) = (0, -2)$ m/s²

3. Keep x-quantities in the x description and y-quantities in the y description while using one shared elapsed time.

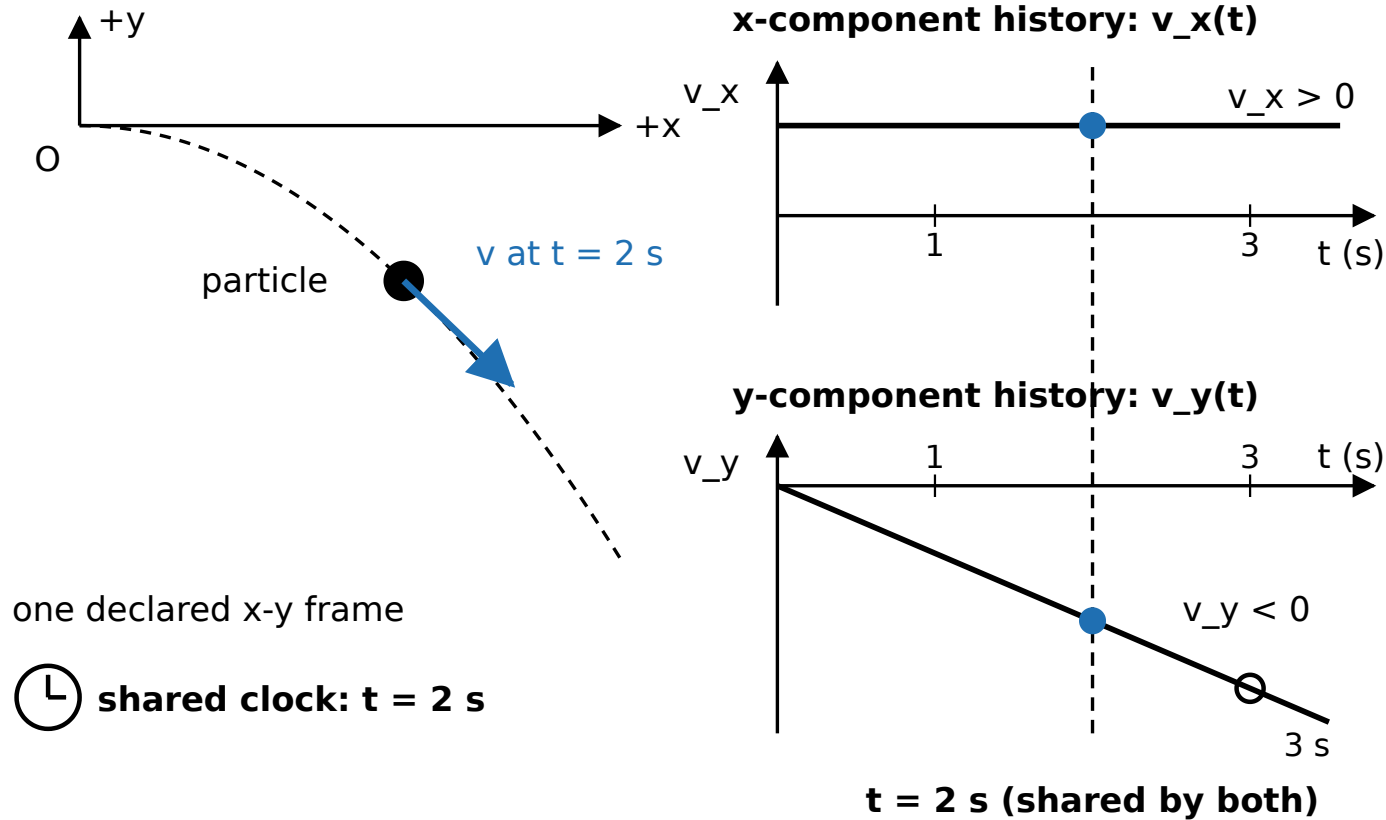
Why valid: The perpendicular component equations are independent in Cartesian kinematics, but both describe the same object during the same physical interval.

Result: $x(1) = 3 \times 1 = 3$ m uses only x-quantities; $y(1) = 5 \times 1 - 1^2 = 4$ m uses only y-quantities; both use $t = 1$ s

4. Recombine only component values that refer to the same frame and the same instant.

Why valid: Mixing x at one time with y at another does not describe any single physical state of the object.

Result: one physically valid plane state reconstructed from simultaneous components



✓ $(v_x(2\text{ s}), v_y(2\text{ s}))$: simultaneous, one valid plane state

✗ $(v_x(2\text{ s}), v_y(3\text{ s}))$: non-simultaneous, rejected (no single instant)

Keep independently evolving x and y descriptions attached to one physical object, one declared frame and one shared clock, so only simultaneous component values are recombined into a plane state.

Worked example

A drone flies in a frame with origin at its launch pad, $+x$ east, $+y$ up and $t = 0$ at launch. Its position is $x(t) = 6t$ m and $y(t) = (8t - t^2)$ m. Find its position at $t = 2$ s, and decide whether the pair $x(2 \text{ s}), y(4 \text{ s})$ is a position the drone ever occupies.

1. The frame, origin and clock are declared once for the whole flight, so both functions share one t .
2. At $t = 2$ s: $x(2) = 6 \times 2 = 12$ m and $y(2) = 8 \times 2 - 2^2 = 12$ m, so the drone is at (12, 12) m.
3. $y(4) = 8 \times 4 - 4^2 = 16$ m, so the pair $x(2 \text{ s}), y(4 \text{ s})$ is (12, 16) m.
4. $x = 12$ m happens only at $t = 2$ s, when y is 12 m rather than 16 m, so (12, 16) m is never a position of the drone.

At $t = 2$ s the drone is at (12, 12) m. The pair (12, 16) m mixes two instants and is not a position the drone ever occupies.

A tempting wrong path

- Independent x and y motions may use different elapsed times.
- The x -motion and y -motion are two different objects that can be paired arbitrarily later.

Diagnose

- At one instant 2 s after launch, may the x equation use $t=2$ s while the y equation uses another time?
- May $x(2\text{ s})$ be combined with $y(3\text{ s})$ to describe the particle's position?

Repair

- No. Independence separates the axis equations; it does not give the same object two clocks.
- No. A plane state is a simultaneous pair. Recombine $x(t)$ and $y(t)$ only at the same t in the same frame.

Check it independently

- Substitute the reported t back into both component functions: each must return its reported coordinate. If one coordinate only matches at a different t , the pair is not a position.
- Reverse the declared $+y$ direction: every y -quantity must change sign together while no x -quantity changes, and the physical position stays the same.

Exit task

A particle has x -velocity 6 m/s and y -velocity -8 m/s at $t=3$ s. Explain what 'independent components' means and what must remain shared between the x and y descriptions.

Model answer

The x and y component equations are solved separately in the same frame, but both describe the same particle at the same $t=3$ s.

1. Keep x quantities in the x equation and y quantities in the y equation.
2. Use one common elapsed time for the same physical instant.
3. Recombine components only after each axis is handled consistently.

Apply constant-acceleration kinematics component by component

You need

- Can use the existing one-dimensional constant-acceleration model.
- Can keep x/y components separate with one common time.
- [Represent one plane-motion event as independent signed x/y component histories that share one elapsed time.](#)
- Derive the constant-acceleration equations from one velocity-time graph (Physics)

The key step

The one-dimensional constant-acceleration equations can be applied to each perpendicular component independently when a_x and a_y are each constant over the same interval.

Check on each axis that its acceleration component is constant, then run the one-dimensional equations once per axis with one shared time.

1. **Check separately whether a_x and a_y are constant over the interval before choosing constant-acceleration equations on either axis.**

Why valid: The familiar one-dimensional kinematic relations are exact only on an axis whose acceleration component is constant over the interval.

Result: per-axis model-validity decision

2. **Apply the one-dimensional constant-acceleration relations independently to x and y using only that axis's signed values and the same t .**

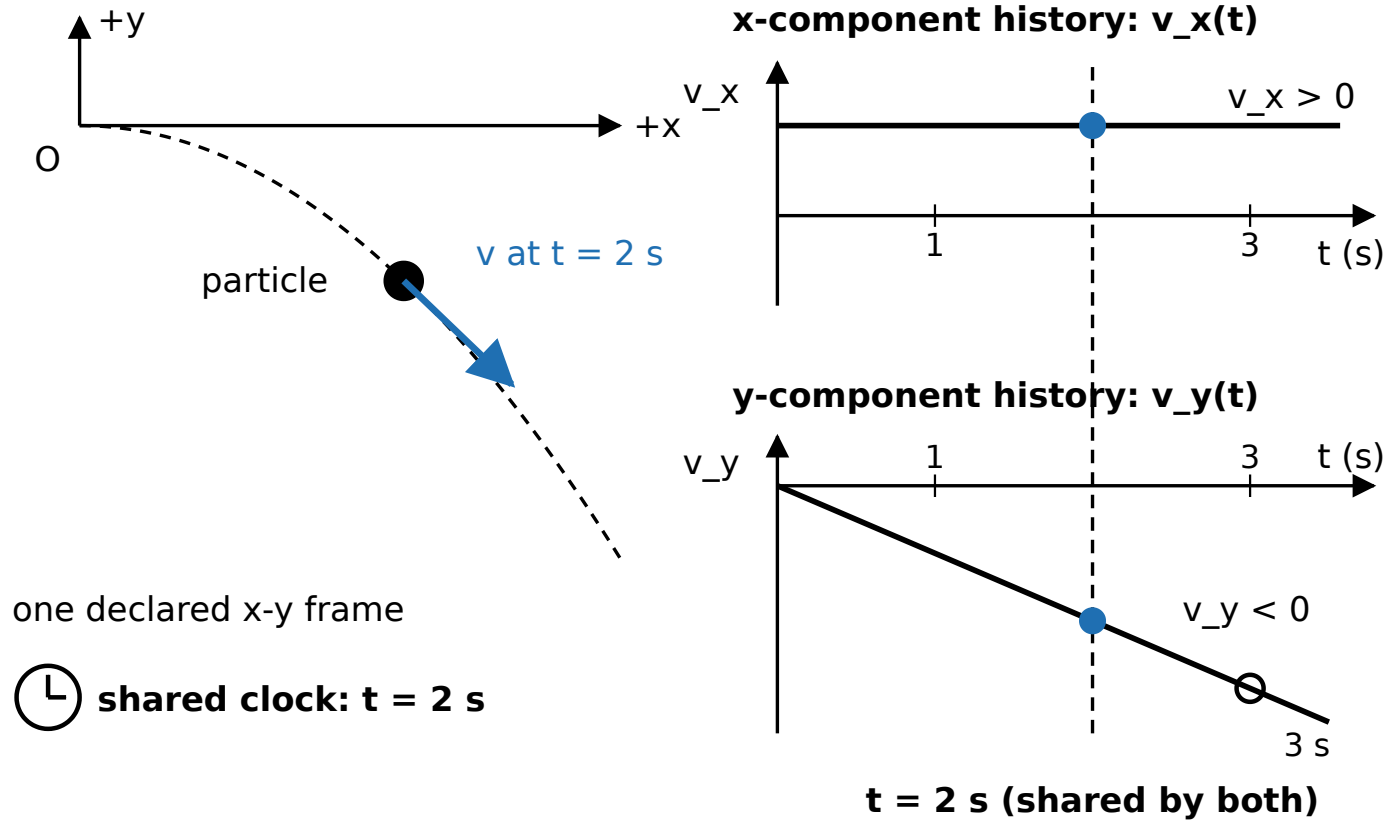
Why valid: Each Cartesian component obeys the same one-dimensional relation when its acceleration component is constant.

Result: $v_x = u_x + a_x t$; $v_y = u_y + a_y t$; $\Delta x = u_x t + 0.5 a_x t^2$; $\Delta y = u_y t + 0.5 a_y t^2$

3. **If one axis determines an event time, reuse that same time on the other axis before recombining and checking the plane state.**

Why valid: The event occurs once for the object; a time obtained from one component is the clock value for every component at that same event.

Result: event time transferred across axes, then one checked plane state



✓ $(v_x(2\text{ s}), v_y(2\text{ s}))$: simultaneous, one valid plane state

✗ $(v_x(2\text{ s}), v_y(3\text{ s}))$: non-simultaneous, rejected (no single instant)

Keep independently evolving x and y descriptions attached to one physical object, one declared frame and one shared clock, so only simultaneous component values are recombined into a plane state.

Worked example

A puck on a tilted air table starts at the origin with velocity $(4 \mathbf{i} + 6 \mathbf{j})$ m/s, and its acceleration is $(0 \mathbf{i} - 2 \mathbf{j})$ m/s² throughout. Using the table's x - y frame, decide which constant-acceleration equations apply on each axis, find when the puck is next at $y = 0$, and find its x -position and velocity at that instant.

1. $a_x = 0$ and $a_y = -2$ m/s² are both constant, so the constant-acceleration relations are exact on both axes.
2. On y : $\Delta y = 6t - 0.5 \times 2 \times t^2 = 6t - t^2$. Setting $\Delta y = 0$ gives $t(6 - t) = 0$, and the later root is $t = 6$ s.
3. On x , at the same $t = 6$ s: $\Delta x = 4 \times 6 = 24$ m.
4. Velocity at $t = 6$ s: $v_x = 4$ m/s and $v_y = 6 - 2 \times 6 = -6$ m/s, so $\mathbf{v} = (4 \mathbf{i} - 6 \mathbf{j})$ m/s.

The puck is next at $y = 0$ at $t = 6$ s, at $x = 24$ m, moving with $\mathbf{v} = (4 \mathbf{i} - 6 \mathbf{j})$ m/s.

A tempting wrong path

- The magnitude of acceleration should be inserted unchanged into both x and y equations.
- If one acceleration component is constant, the constant-acceleration equations are automatically valid on both axes.
- A time found from the x equation may be replaced by a different convenient time in the y equation.

Diagnose

- If $a = (10 \mathbf{i} + 4 \mathbf{j}) \text{ m/s}^2$, should both component equations use $a = \sqrt{116}$?
- If a_x is constant but a_y changes with time, may the same constant-acceleration displacement formula be used exactly on y over the whole interval?
- If x -motion says an event occurs at $t = 3 \text{ s}$, what time belongs in the y equation for that same event?

Repair

- No. Use $a_x = 10$ only in the x equation and $a_y = 4$ only in the y equation; the magnitude is not either component.
- No. Model validity is checked per axis. A constant a_x does not make a_y constant.
- Use $t = 3 \text{ s}$. Both components describe the same event and therefore share one clock.

Check it independently

- Every term of $\Delta y = u_y t + 0.5 a_y t^2$ must come out in metres: $(\text{m/s})(\text{s})$ and $(\text{m/s}^2)(\text{s}^2)$. A velocity term in m/s beside them means a t is missing.
- Set $a_x = 0$: the x equations must reduce to uniform motion, $v_x = u_x$ and $\Delta x = u_x t$, whatever happens on y .

Exit task

A particle starts with $u=(0,5)$ m/s and constant $a=(10,4)$ m/s². Set up the x and y position equations at common time t without solving a particular numerical t.

Model answer

Use $\Delta x = 0 \cdot t + 0.5(10)t^2$ and $\Delta y = 5t + 0.5(4)t^2$ with the same t.

1. Read signed component initial velocities and accelerations.
2. Apply the 1D constant-acceleration displacement relation separately.
3. Keep one common elapsed time.

Select the gravity-only projectile model before using projectile consequences

You need

- Can solve 2D constant-acceleration component equations.
- If an oblique launch angle is given, angle-based vector decomposition is available as a separate prerequisite for that question.
- [Apply constant-acceleration kinematics independently to x and y components of one plane-motion event.](#)

The key step

After release in the ideal near-Earth model, projectile motion is not a new set of laws: it is the same component kinematics with $a_x=0$ and $a_y=-g$ (for +y upward).

Decide from what acts after release whether the gravity-only model applies, and only then put $a_x = 0$ and $a_y = -g$ into the component equations.

1. **Decide whether the object is in ideal near-Earth free flight after release; neglect air resistance, declare +y, then set $a_x=0$ and $a_y=-g$.**

Why valid: The standard projectile equations are a conditional specialization of 2D constant acceleration, not an automatic label for every launched object.

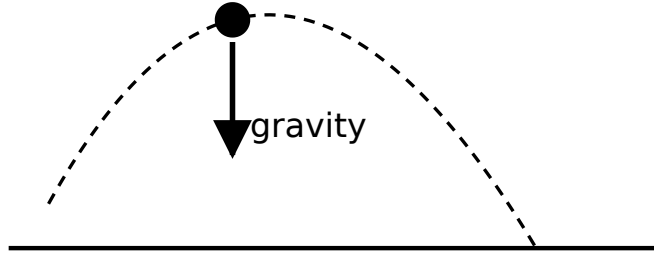
Result: gravity-only projectile acceleration state

2. **Use supplied or separately resolved u_x and u_y with one common t in the horizontal and vertical component equations.**

Why valid: Horizontal, oblique and downward/upward initial components change the initial state, not the gravity-only model.

Result: $v_x=u_x$; $v_y=u_y-g t$; $\Delta x=u_x t$; $\Delta y=u_y t-0.5g t^2$

A: launched, then free flight



After release, acting on the object:

- gravity (Earth on object), down
- drag: neglected

Gravity only? yes

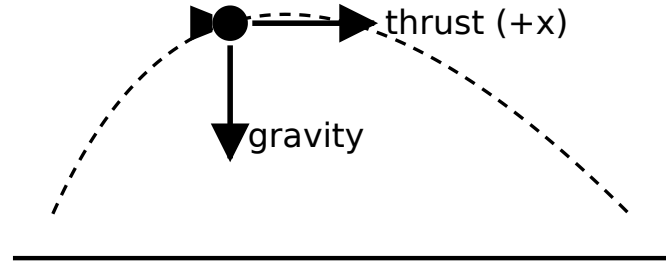
$$a_x = 0$$

$$a_y = -g \text{ (+y up)}$$

Ideal projectile model: VALID

$a_x = 0$ and $a_y = -g$ both hold after release.

B: launched, motor stays on



After release, acting on the object:

- gravity (Earth on object), down
- thrust (motor on object), horizontal
- drag: neglected

Gravity only? no (thrust retained)

$$a_x \neq 0 \text{ (from the thrust)}$$

$$a_y = -g \text{ (+y up)}$$

Ideal projectile model: REJECTED

Active thrust violates $a_x = 0$.
Use the general 2D
constant-acceleration model.

Make the ideal projectile specialization conditional on the interactions that remain after release: gravity-only free flight licenses $a_x=0$ and $a_y=-g$; retained thrust, significant drag or contact requires a more general model.

Worked example

A ball is thrown from a balcony with $u_x = 8$ m/s and $u_y = 6$ m/s. After release only gravity acts, and air resistance is negligible. Take +y upward and $g = 10$ m/s². Decide whether the ideal projectile model applies, write its component relations, and find the ball's displacement and velocity 1 s after release.

1. After release the only retained interaction is gravity, so the ideal model applies: $a_x = 0$ and $a_y = -10$ m/s².
2. Component relations: $v_x = 8$, $v_y = 6 - 10t$, $\Delta x = 8t$ and $\Delta y = 6t - 5t^2$, in metres and seconds.
3. At $t = 1$ s: $\Delta x = 8$ m and $\Delta y = 6 - 5 = 1$ m, so the displacement is $(8 \mathbf{i} + 1 \mathbf{j})$ m.
4. At $t = 1$ s: $v_x = 8$ m/s and $v_y = 6 - 10 = -4$ m/s, so $\mathbf{v} = (8 \mathbf{i} - 4 \mathbf{j})$ m/s.

The model applies with $\mathbf{a} = (0, -10)$ m/s². After 1 s the displacement is $(8 \mathbf{i} + 1 \mathbf{j})$ m and the velocity is $(8 \mathbf{i} - 4 \mathbf{j})$ m/s.

A tempting wrong path

- A projectile needs a horizontal force to keep moving horizontally.
- A horizontal launch has no vertical velocity throughout the flight.

Diagnose

- If $a_x=0$ after release, can v_x still be nonzero?
- One second after a horizontal launch, is v_y still zero under gravity?

Repair

- Yes. Zero horizontal acceleration means horizontal velocity stays constant, not that it becomes zero.
- No. $u_y=0$ only at release; $a_y=-g$ makes $v_y=-gt$ afterward.

Check it independently

- Set $g = 0$: the equations must reduce to straight-line uniform motion with the launch velocity (u_x , u_y).
- With $+y$ up, v_y must fall by g every second whatever the launch direction; a v_y that grows while the object rises means the sign of g is wrong.

Name the requested event as a component condition before solving for its time:
 $v_y = 0$ at the apex, $\Delta y = 0$ at a return to launch height.

1. **Translate the requested event into the component condition that defines it before solving for time.**

Why valid: Apex, same-height return, ground impact and a specified later instant are different events;

each is identified by a different component condition.

Result: event condition chosen before calculation

2. **For an apex event, set $v_y=0$ while keeping $v_x=u_x$ and $a_y=-g$.**

Why valid: At the highest point only the vertical velocity component is momentarily zero; gravity still accelerates downward and horizontal motion continues.

Result: apex condition $v_y=0$ with nonzero horizontal velocity and nonzero downward acceleration

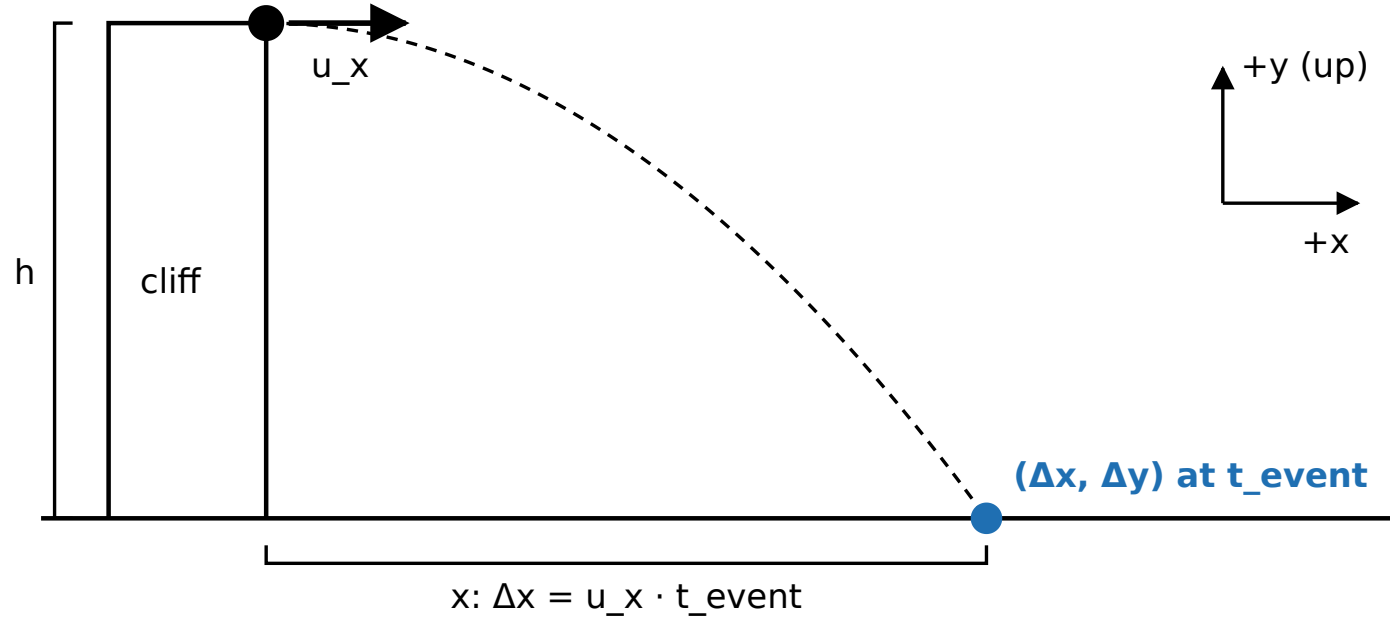
3. **For a return to the launch height, set $\Delta y=0$ and treat any same-height flight-time/range shortcut as conditional on that equality.**

Why valid: The shortcut follows from the vertical displacement returning to its initial value; it is not valid when the landing height differs.

Result: same-height event condition and its validity boundary

Requested event: the stone reaches the ground

event condition (y): $\Delta y = -h$



one shared event timeline

event time t_{event}



y: $\Delta y = -h$ at the same t_{event}

The one t_{event} is reused unchanged on x and on y.

Stage 1 of 4

Make event selection explicit: one component condition identifies the requested event time, and that same time is then reused on every component before reconstructing the requested state.

Worked example

An ideal projectile is launched from level ground with $u_x = 6$ m/s and $u_y = 15$ m/s. Take +y upward and $g = 10$ m/s². Name the component condition for each event, then find the time and height of the apex, and the time and range when it lands back at launch height.

1. Apex: the condition is $v_y = 0$, while v_x stays 6 m/s and a_y stays -10 m/s². $15 - 10t = 0$ gives $t = 1.5$ s.
2. Apex height: $\Delta y = 15 \times 1.5 - 5 \times 1.5^2 = 22.5 - 11.25 = 11.25$ m.
3. Landing at launch height: the condition is $\Delta y = 0$. $15t - 5t^2 = 0$ gives $t = 0$ (the launch) or $t = 3$ s, so the landing is at $t = 3$ s.
4. Range at the same $t = 3$ s: $\Delta x = 6 \times 3 = 18$ m.

Apex at $t = 1.5$ s, 11.25 m above launch, with velocity $(6 \mathbf{i} + 0 \mathbf{j})$ m/s; landing at $t = 3$ s, 18 m from launch.

A tempting wrong path

- At the highest point the whole velocity and the acceleration are both zero.
- At two points of the same height on an ideal trajectory, the velocity vectors are identical.

Diagnose

- At the apex of an oblique projectile, which of v_x , v_y and a_y is zero?
- On ascent and descent at the same height, how do the horizontal and vertical velocity components compare?

Repair

- Only v_y is zero at that instant. v_x remains u_x and a_y remains $-g$.
- v_x is the same, while v_y has equal magnitude and opposite sign; the speed matches but the velocity vector does not.

Check it independently

- For a return to launch height the flight time must be exactly twice the apex time, $2u_y/g$ against u_y/g , because the rise and the fall mirror each other.
- Set $u_y = 0$ in the apex condition: the apex must fall at $t = 0$, as it does for a launch that starts level.

Write the landing event from the actual drop while keeping the actual u_y , then read every requested quantity at that one event time.

1. **For any unequal-height landing, use the actual $\Delta y = y_f - y_i$ in the vertical position equation and retain the actual u_y ; for a horizontal launch specifically, set $u_y = 0$, then**

obtain the event time vertically and share it with x.

Why valid: Initial vertical velocity and landing displacement describe different facts. The landing event is fixed by the actual vertical geometry, while $u_y=0$ is only the horizontal-launch initial condition.

Result: $\Delta y = y_f - y_i = u_y t_{\text{event}} - 0.5g t_{\text{event}}^2$; for horizontal launch $u_y=0$; reuse the solved t_{event} on x

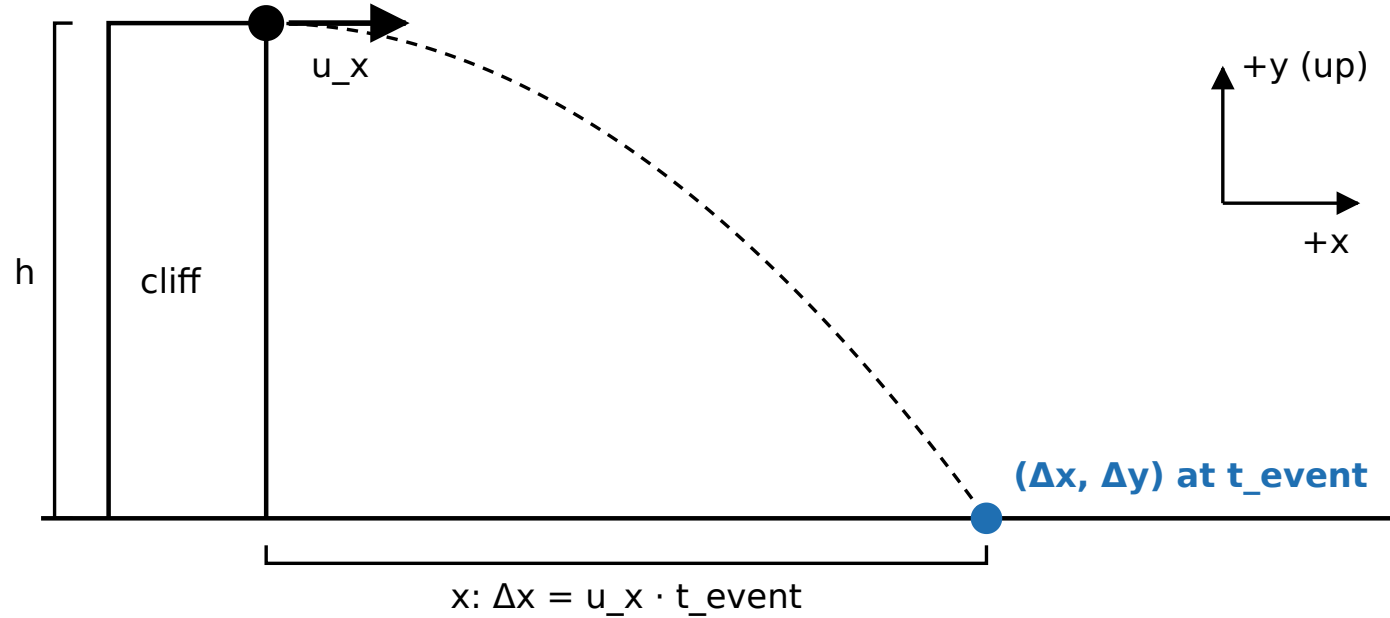
2. At the selected event time, reconstruct the requested velocity, speed, direction, displacement or range from simultaneous components and check qualitative limits.

Why valid: The reported projectile result is a consequence of one component model and one event, so its components must refer to the same instant and satisfy $a_x=0$, $a_y=-g$.

Result: checked projectile consequence at one physical event

Requested event: the stone reaches the ground

event condition (y): $\Delta y = -h$



one shared event timeline

event time t_{event}



y: $\Delta y = -h$ at the same t_{event}

The one t_{event} is reused unchanged on x and on y.

Stage 1 of 4

Make event selection explicit: one component condition identifies the requested event time, and that same time is then reused on every component before reconstructing the requested state.

Worked example

A ball is thrown from the edge of a roof 25 m above level ground with $u_x = 5$ m/s and $u_y = 20$ m/s. Take +y upward and $g = 10$ m/s², and neglect air resistance. Find when it hits the ground, how far from the building it lands, and its impact velocity and speed.

1. The landing event is $\Delta y = -25$ m, because the ground is below the launch point, and $u_y = 20$ m/s is kept: $-25 = 20t - 5t^2$.
2. Rearranged, $5t^2 - 20t - 25 = 0$, so $t^2 - 4t - 5 = 0$ and $(t - 5)(t + 1) = 0$. The physical root is $t = 5$ s.
3. Reuse $t = 5$ s horizontally: $\Delta x = 5 \times 5 = 25$ m.
4. At $t = 5$ s: $v_x = 5$ m/s and $v_y = 20 - 10 \times 5 = -30$ m/s, so $\mathbf{v} = (5 \mathbf{i} - 30 \mathbf{j})$ m/s and the speed is $\sqrt{(5^2 + 30^2)} = \sqrt{925} \approx 30.4$ m/s.

Impact at $t = 5$ s, 25 m from the building, with $\mathbf{v} = (5 \mathbf{i} - 30 \mathbf{j})$ m/s and a speed of about 30.4 m/s.

A tempting wrong path

- A horizontal launch has no vertical velocity throughout the flight.
- Same-height flight formulas may be used whenever the question is about a projectile range or flight time.

- Horizontal speed determines how long a horizontally launched object takes to fall.

Diagnose

- One second after a horizontal launch, is v_y still zero under gravity?
- A ball launched from a roof lands on the ground below. May the same-height time-of-flight shortcut be used without checking the vertical displacement?
- If the horizontal launch speed doubles from the same height, does the ideal fall time change?

Repair

- No. $u_y=0$ only at release; $a_y=-g$ makes $v_y=-gt$ afterward.
- No. Same-height shortcuts require $\Delta y=0$ between launch and landing. For unequal heights, solve the vertical position condition.
- No. The vertical position equation determines the fall time; horizontal speed changes the range, not the fall time.

Check it independently

- The impact speed from the components must match $v^2 = u^2 + 2gh$ for a drop h , a route that uses no time at all.
- Let the drop shrink to zero: the landing time must reduce to the same-height value $2u_y/g$.

Exit task

A stone is launched horizontally from a cliff with speed u . With $+y$ upward and air resistance neglected, write its component acceleration, velocity and displacement relations after release and explain why one common t is used.

Model answer

$a_x=0$, $a_y=-g$; $v_x=u$, $v_y=-gt$; $\Delta x=ut$, $\Delta y=-0.5gt^2$, all at the same elapsed time t .

1. After release only gravity acts in the ideal model.
2. Gravity has no horizontal component in the chosen frame.
3. Both component coordinates describe the same stone at the same instant.